Question 4: Java program to count the number of characters in each word using Hadoop MapReduce

each word using Hadoop MapReduce

Aim

To count the number of characters present in each word in an input file using Hadoop MapReduce.

Procedure

1. Create an input text file containing words.

2. The mapper reads each line, splits it into words, and emits the word as key and its character count as value.

3. The reducer receives each word and its count and writes the result.

4. Compile the Java program, create a JAR file, upload the input to HDFS and run the job.

5. Display the output using hdfs dfs -cat.

Program

```
import java.io.IOException;

import org.apache.hadoop.conf.Configuration;

import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.*;

import org.apache.hadoop.mapreduce.*;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;

import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCharacterCount {

public static class M extends Mapper<LongWritable,Text,Text,IntWritable> {

public void map(LongWritable key, Text value, Context c)

throws IOException, InterruptedException {

String line = value.toString().toLowerCase();

for (String word : line.split("\\W+")) {

if (!word.isEmpty())

c.write(new Text(word), new IntWritable(word.length()));

}

}

}
```

```
public static class R extends Reducer<Text,IntWritable,Text,IntWritable> {
public void reduce(Text word, Iterable<IntWritable> counts, Context c)
throws IOException, InterruptedException {
// All occurrences of the same word have the same character length.
int length = 0;
for (IntWritable x : counts) length = x.get();
c.write(word, new IntWritable(length));
}
}
public static void main(String[] args) throws Exception {
Job job = Job.getInstance(new Configuration(), "Characters in Each Word");
job.setJarByClass(WordCharacterCount.class);
job.setMapperClass(M.class);
job.setReducerClass(R.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));
System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

Compile and execute

```
export HADOOP_CLASSPATH=$(hadoop classpath)
javac -classpath "$HADOOP_CLASSPATH" -d classes WordCharacterCount.java
jar -cvf wordchars.jar -C classes .
hdfs dfs -mkdir -p /input/words
hdfs dfs -put words.txt /input/words/
hadoop jar wordchars.jar WordCharacterCount /input/words /output/wordchars
hdfs dfs -cat /output/wordchars/part-r-00000
```

Output of Program

```
bigdata 7
```

hello 5

hadoop 6

world 5

Result

The MapReduce program successfully displays the character length of each distinct word. Repeated occurrences of a

word are grouped under the same key.